Project Initialization and Planning Phase

Date	12 July 2026
Team ID	XXXXXX
Project Title	Credit Card Approval Prediction
Maximum Marks	5 Marks

Project Proposal (Proposed Solution) report

The proposed solution aims to automate the credit card approval process using machine learning techniques. The system analyzes applicant information and predicts whether a credit card application should be approved or rejected. This improves decision-making accuracy, reduces manual effort, minimizes financial risk, and provides faster service to customers.

Project Overview	
Objective	The primary objective is to develop a machine learning-based system that predicts credit card approval accurately and efficiently, reducing manual verification time and improving decision-making.
Scope	The project focuses on analyzing applicant data, training machine learning models, and providing instant approval predictions through a user-friendly web application.
Problem Statement	
Description	Traditional credit card approval processes are time-consuming and prone to human errors. Manual verification may result in inconsistent decisions and delayed approvals.
Impact	An automated prediction system improves operational efficiency, reduces financial risk, speeds up approval decisions, and enhances customer satisfaction
Proposed Solution	
Approach	Develop a machine learning model using applicant information such as income, employment status, credit history, and other financial details to predict credit card approval.
Key Features	Machine Learning-based approval prediction Fast and accurate decision-making User-friendly web interface

	Reduced manual verification Improved consistency in approval decisions Secure handling of applicant data

Resource Requirements

Resource Type	**Description**	**Specification/Allocation**
Hardware		
Computing Resources	CPU/GPU specifications, number of cores	Intel i5 Processor (or higher)
Memory	RAM specifications	8 GB RAM
Storage	Disk space for data, models, and logs	512 GB SSD
Software		
Frameworks	Python frameworks	Flask
Libraries	Additional libraries	Scikit-learn, Pandas, NumPy, Matplotlib, Pickle
Development Environment	IDE	Jupyter Notebook, Visual Studio Code
Data		
Data	Source, size, format	Kaggle Credit Card Approval Dataset, CSV format